EE-170L Computer Programming Lab

Lab Session 01

Installation and Setup of C++ on Microsoft Windows using Visual Studio Code

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1. Objective

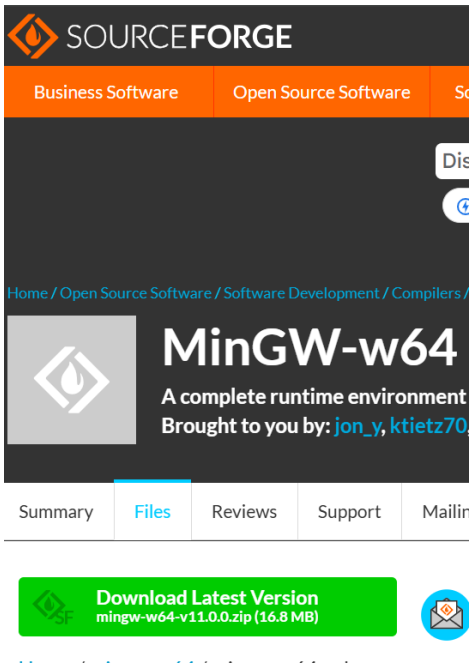
The objective of this lab session was to install and configure the C++ programming environment on a Microsoft Windows system using Visual Studio Code (VS Code) along with the MinGW compiler. The purpose was to ensure that the system is capable of writing, compiling, and executing C++ programs successfully.

2. Download and Install Visual Studio Code

- Downloaded Visual Studio Code from the official website: https://code.visualstudio.com/Download.
- Clicked on the 'Download for Windows' option.
- Executed the downloaded installer file.
- Followed the on-screen instructions to complete the installation.

3. Download and Install MinGW Compiler

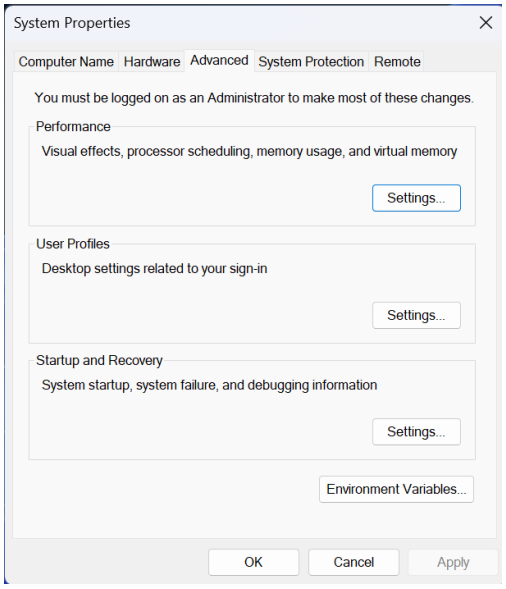
- Visited the MinGW-w64 website: https://sourceforge.net/projects/mingw/.
- Downloaded the installer according to the system architecture (32-bit or 64-bit).
- Ran the installer and completed the setup process.
- Opened the MinGW installation manager.
- Selected the package 'mingw32-gcc-g++'.
- Clicked on 'Install' and then selected 'Apply Changes' to finalize installation.



4. Add g++ Path to System Environment Variables

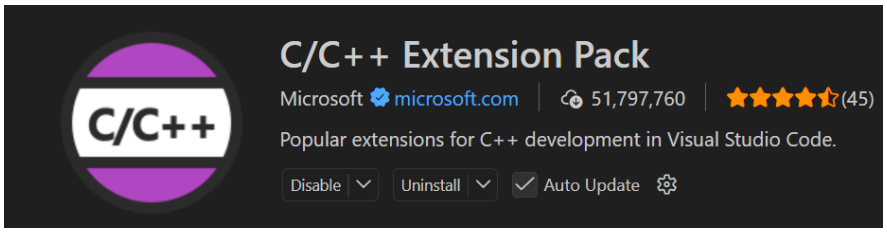
- Located the MinGW installation directory (e.g., C:/MinGW/bin).
- Copied the path of the 'bin' folder.
- Opened the Start Menu and searched for 'Environment Variables'.
- Selected 'Edit the system environment variables'.
- Clicked on 'Environment Variables' in the System Properties window.
- Under 'System Variables', selected 'Path' and clicked 'Edit'.
- Clicked 'New' and pasted the MinGW bin path.

- Clicked 'OK' to save all changes.



5. Install Required Extensions in Visual Studio Code

- Opened Visual Studio Code.
- Clicked on the Extensions icon from the left sidebar.
- Searched for 'C++' and installed the 'C/C++' extension by Microsoft.
- Restarted VS Code after installation.
- Searched for 'Code Runner' and installed the 'Code Runner' extension by Jun Han.
- Restarted VS Code to complete the setup.



6. Conclusion

In this lab session, the C++ development environment was successfully installed and configured using Visual Studio Code and the MinGW compiler. All necessary tools and extensions were properly set up, enabling smooth compilation and execution of C++ programs. This environment will be used for future laboratory experiments throughout the semester.

7. Useful video link

https://youtu.be/VTLCoHnyACE?si=mNzUzGG2A5N6P_BK